

Understanding Skin Dynamics

Force and Displacement are two key parameters for transducer selection during device design. Here we report both force and displacement levels at detection threshold levels for 10-500 Hz cutaneous stimulations using two types of holds, i.e., stylus-hold, and ball/puck-hold. In addition, the mechanical impedance is calculated and discussed in the context of device design.

Notable Publications:

- Israr, A., Choi, S., & Tan, H. Z. (2007, March). Mechanical impedance of the hand holding a spherical tool at threshold and suprathreshold stimulation levels. In Second Joint EuroHaptics Conference and Symposium on Haptic Interfaces for Virtual Environment and Teleoperator Systems (WHC'07) (pp. 56-60). IEEE. [\[Link\]](#)
- Israr, A., Choi, S., & Tan, H. Z. (2006, October). Detection threshold and mechanical impedance of the hand in a pen-hold posture. In 2006 IEEE/RSJ international conference on intelligent robots and systems (pp. 472-477). IEEE. [\[Link\]](#)

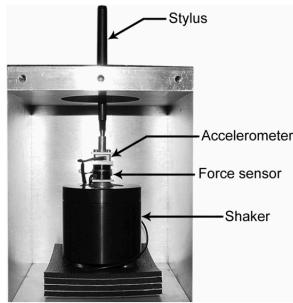


Figure 1. Shaker assembly with one of the side panels removed.

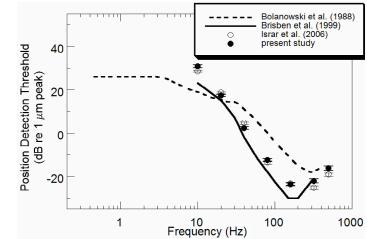


Fig. 3. Measured position detection thresholds and comparison with published data.

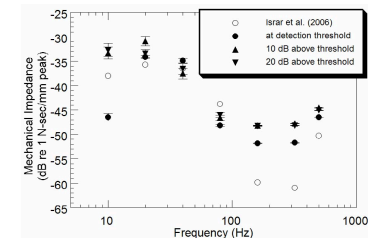


Fig. 5. Mechanical impedance of the hand at threshold and supra-threshold levels

In addition, we measure the sensitivity of the finger for mixed reality force feedback.

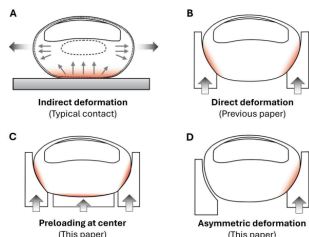


Fig. 3. (A) Skin deformation at the fingertips edges caused by normal contact force. (B) Principle of edge deformation presented in our previous study [10]. (C) Edge deformation with an additional central preload examined in this study. (D) Principle of asymmetric edge deformation investigated in this study.

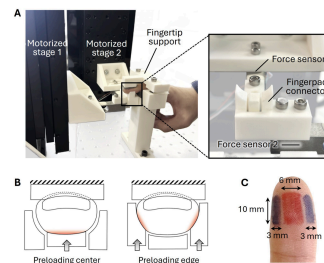


Fig. 4. (A) Experimental setup for the user perception test. (B) Mechanisms of deformation generation at the center and the edges of the fingertip. (C) Contact area on the fingertip. (All images from [10])

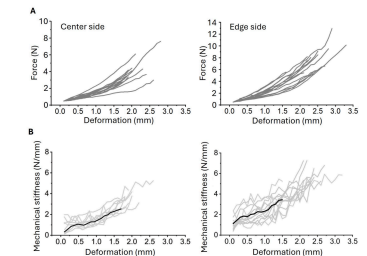


Fig. 6. (A) Recorded relationship between skin deformation and contact force at the center and edges of the fingertip for all participants. (B) Calculated mechanical stiffness as a function of skin deformation at the center and edges of the fingertip. The gray lines represent individual participant data, while the black line represents the averaged data.

- Youn, J. H., Chen, Y., Israr, A., & Shultz, C. (2026). Sensitivity at the Edges of the Finger Enables Mixed Reality Force Feedback. IEEE Transactions on Haptics. [\[Link\]](#)

A psychophysical model of motorcycle handlebar vibrations:

We developed a perception-based quantitative model to relate broadband vibrations transmitted through a motorcycle handlebar to a rider's hands. The test apparatus consisted of the handlebar of a motorcycle rig assembly driven by a computer-controlled actuator. Participants were instructed to hold the handlebar and maintain a sitting posture as they would while riding a motorcycle. The best predictive model, in the sense that the estimated total energy correlated linearly with the subjective magnitude ratings obtained in Exp. 2, were based on the detection threshold data obtained in Exp. 1. Specifically, the strength of each vibration component was calculated relative to the human detection threshold at the same frequency, thereby taking into account human sensitivity to vibration signals at different frequencies.

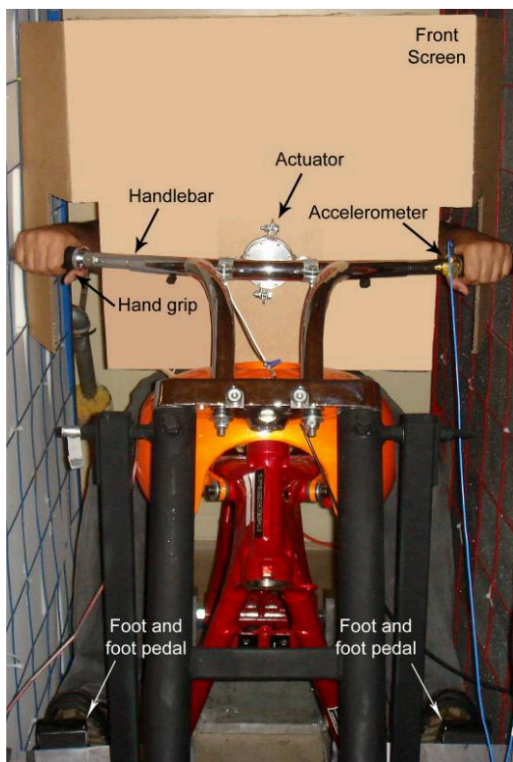


Figure 3. Experimental setup with a participant holding the handlebar.

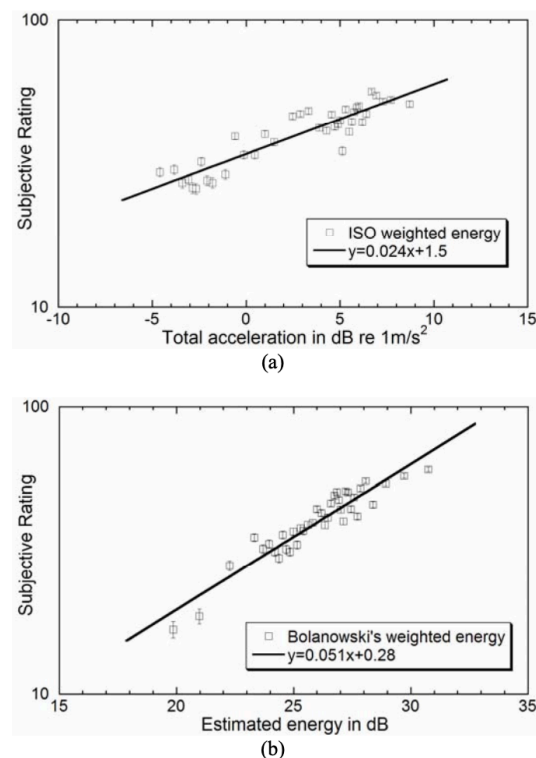


Figure 6. Normalized subjective rating vs. total perceived acceleration in dB by (a) ISO weighted function and (b) Bolowski's detection thresholds.

- Israr, A., Tan, H. Z., Mynderse, J., & Chiu, G. T. C. (2007, January). A psychophysical model of motorcycle handlebar vibrations. In ASME International Mechanical Engineering Congress and Exposition (Vol. 43033, pp. 1233-1239). [\[Link\]](#)